MySQL 8.0 Reference Manual

Version 8.0 Features and Functionality

1. Authentication and User Management

MySQL 8.0 uses the caching_sha2_password as the default authentication plugin, which provides more secure password encryption than the previous mysql_native_password plugin:

```sql
-- Default behavior now uses caching_sha2_password
CREATE USER 'username'@'host' IDENTIFIED BY 'password';

-- To use legacy authentication:
CREATE USER 'username'@'host' IDENTIFIED WITH mysql_native_password BY 'password';
```

Password validation is built-in and enabled by default through the validate_password component:

```sql
INSTALL COMPONENT 'file://component_validate_password';
```

2. Important System Variables

Variable Name	Default	Description
max_connections	151	Maximum permitted number of client connections
innodb_buffer_pool_size	128M	Size of the buffer pool for caching data and indexes
cte_max_recursion_depth	1000	Maximum recursion depth for common table expressions
transaction_isolation	REPEATABLE-READ	Default transaction isolation level

The query cache has been removed in MySQL 8.0. Variables like query_cache_size and query_cache_type no longer exist.

3. SQL Syntax Elements

Window Functions

MySQL 8.0 introduces SQL window functions:

```sql
SELECT
    name,
    department,
    salary,
    RANK() OVER (PARTITION BY department ORDER BY salary DESC) as dept_rank
FROM employees;
```

Common Table Expressions (CTEs)

Recursive and non-recursive CTEs are now supported:

```sql
WITH RECURSIVE cte (n) AS (
    SELECT 1
    UNION ALL
    SELECT n + 1 FROM cte WHERE n < 5
)
SELECT * FROM cte;
```

JSON Enhancements

New functions for JSON including JSON_TABLE:

```sql
SELECT *
FROM JSON_TABLE(
    '{"employees": [{"name": "John", "age": 30}, {"name": "Jane", "age": 25}]}',
    '$.employees[*]' COLUMNS(
        name VARCHAR(20) PATH '$.name',
        age INT PATH '$.age'
    )
) AS employee_table;
```

4. Removed Features

The following features have been removed in MySQL 8.0:

- Query cache (completely removed)
- The mysql_install_db script (use mysqld --initialize instead)
- ENCRYPT(), PASSWORD(), OLD_PASSWORD() functions
- db, host, and user grant tables (user privileges are now handled through role-based access)
- The DATA DIRECTORY and INDEX DIRECTORY table options for MyISAM tables

- The IGNORE clause in DROP DATABASE statements

5. InnoDB Enhancements

- Instant ADD COLUMN for InnoDB tables
- Default REDO log encryption
- InnoDB cluster enhancements
- Improved crash recovery performance
- InnoDB now supports Data Dictionary that stores metadata in InnoDB tables

6. Error Codes

New error codes in MySQL 8.0:

- 3100: Missing JSON key in value
- 3140: Invalid JSON path expression
- 3546: Lock wait timeout exceeded
- 3930: Component cannot be installed
- 4024: Removed features/functionality error

7. Replication

MySQL 8.0 enhancements for replication:

- Binary logs are encrypted by default
- Binary log transaction compression
- JSON binary format improvements
- Default replication format is now ROW instead of STATEMENT
- Improved Group Replication
- Improved replication performance with write set-based conflict detection

8. Character Set and Collation

- The default character set has changed from latin1 to utf8mb4
- The default collation is now utf8mb4_0900_ai_ci